Bench exercise: one stick, four legs

Japanese Joinery by Hand · Episode 1 · Make a story stick (*bakabō* 馬鹿棒) for a small project and transfer it to four parts. About 45 minutes. No joinery is cut; this is layout only.

You need

- 4 leg blanks for a small stool or side table, about 45 × 45 mm (1¾ in) and a little over the finished length
- 1 straight, plain batten, longer than the legs (scrap is fine: 30–45 mm wide, 10–15 mm thick)
- A square, a sharp pencil or marking knife, and a tape or rule **used once only, on the stick**

Example positions (from the leg's bottom end)

Mark	mm	in	Note on stick
bottom end	0	0	BOTTOM
stretcher, bottom edge	120	4¾	STR
stretcher, top edge	150	5⅞	STR
apron, bottom edge	370	14⅝	APRON
top end	450	17¾	TOP

Use your own project's numbers. These are only an example; inch values are rounded.

1 · Make the stick

1. Square one end of the batten. That end is the stick's reference end.
2. From that end, measure each position **once** and square a line across the stick.
3. Beside every line, write a short note saying what it is for. Mark the side of each line the part goes on.
4. Put the tape away. You will not need it again.

2 · Transfer to four legs

1. Pick one end of each leg as its bottom (reference) end and mark it.
2. Lay the stick against leg 1 with the reference ends exactly flush. Clamp it, or hold it firmly.
3. At each line, square across from the stick onto the leg. Copy each note too.
4. Repeat on legs 2, 3 and 4. Use the same stick, the same reference end, and the same face of the stick against the leg.



3 · Check your work: the payoff

- Stand the four legs side by side with their bottom ends against a straightedge or fence. Lay a long square across all four at each mark: **the lines should run straight across all four legs.**
- For comparison, lay out a fifth blank with the tape, measuring each mark from the last one. Put it beside the others and look at the top marks.
- Remember the point: the stick doesn't make the legs *accurate*, it makes them *identical*. If the stick is 1 mm off your plan, all four legs are 1 mm off together, and the joints still agree.

Common slips

- **Ends not flush.** Any offset at the reference end moves every mark on that leg. Butt both ends against a stop block.
- **The stick turned end for end or flipped.** Write "BOTTOM" big on the stick's reference end.
- **A slipping stick.** Clamp it. A shifted stick makes one leg disagree with the rest.

Optional, toward Episode 2: also strike a center line down one face of each leg and the stick. In Japanese layout, the center line is the main reference (*shinzumi* 芯墨). Marking and cutting tools are taught in later episodes.